



Hafta Epsilon2

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this
license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package com.mycompany.vylab10;
/**
 *
 * @author user
 */
public class SeparateChaining<T extends Comparable> {
    HashNode<T> [] table;
    public SeparateChaining(int size){
        this.table=new HashNode[size];
    }
    //hash fonksiyonu sonucu çıkan index'i döndürür
    public int hashFunction(int key){
        return key%this.table.length;
    }
    public void display(){
        for (int i = 0; i < this.table.length; i++) {
            if(this.table[i]==null)
                System.out.println(this.table[i]);
            else{
                HashNode<T> iterator=this.table[i];

                while(iterator.next!=null){
                    System.out.print(iterator+"-->");
                    iterator=iterator.next;
                }
                System.out.println(iterator);
            }
        }
    }
}
```

```

}
public void insert(int key, T val){
    int index=hashFunction(key);
    HashNode<T> newNode=new HashNode(key, val);
    if(table[index]==null){
        table[index]=newNode;
    }else{
        HashNode<T> iterator=table[index];
        while(iterator.next!=null)
            iterator=iterator.next;
        iterator.next=newNode;
    }
}
public void delete(int key){
    if(!search(key)){
        System.out.println("böyle bir kayıt yok");
        return;
    }else{
        int index=hashFunction(key);
        if(this.table[index].key==key)
            this.table[index]=this.table[index].next;
        else{
            HashNode<T> iterator=this.table[index];
            while(iterator.next.key!=key)
                iterator=iterator.next;
            iterator.next=iterator.next.next;
        }
    }
}
public boolean search(int key){
    int index=hashFunction(key);
    HashNode<T> iterator=this.table[index];
    while(iterator!=null){
        if(iterator.key==key)
            return true;
        iterator=iterator.next;
    }
    return false;
}

```

```

public T accessRecord(int key){
    if(!search(key))
        return null;
    int index=hashFunction(key);
    HashNode<T> iterator=this.table[index];
    while(iterator!=null){
        if(iterator.key==key)
            return iterator.value;
        iterator=iterator.next;
    }
    return null;
}

}

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this
license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package com.mycompany.vylab10;

/**
 *
 * @author user
 */
public class HashNode<T extends Comparable> {
    int key;
    T value;
    HashNode<T> next;
    public HashNode(int key, T val){
        this.key=key;
        this.value=val;
        this.next=null;
    }
    public String toString(){
        return this.key+ " " + this.value;
    }
}

```

```
}

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this
 license
 */
package com.mycompany.vylab10;
/**
 *
 * @author user
 */
public class VYLAB10 {
    public static void main(String[] args) {
        SeparateChaining<String> sp=new SeparateChaining(7);

        sp.insert(3, "ali");
        sp.insert(10, "veli");
        sp.insert(17, "ayse");
        sp.insert(8, "ahmet");
        sp.display();

    }
}
```